



E-Commerce Data Transformation & Data Modelling

Overview:

Business Problem: Analyze product order distributions, category profit margins, and sub-category sales performance to identify key growth drivers and optimize the store's product portfolio.

Objective: This project focuses on organizing, analyzing, from a different type of datasets. The data sets are Sales Target, List of Order, Order Details. Using Power BI tool, the workflow transforms raw datasets into a structured summary.

Dataset Description:

This E-commerce data sets tracks product records across three e-commerce primary categories: Clothing, Electronics, Furniture. Each entry details structural product characteristics, including descriptive item names, specific sub-category classifications, and sales target. For financial and operational reporting, the data maps the sales number of individual items against their category and sub-category. The dataset provides critical operational insights for business dashboard construction, highlights category profit margin and sub-category sales performance to identify key growth and optimize the store's product

Column Description:

List of order Table:

Column Name	Column Description
Order ID	Showing order id
Order Date	Showing order date
Customer Name	Showing customer name
State	Showing the customer state
City	Showing the customer city

Order Details Table:

Column Name	Column Description
Order ID	Showing order id
Amount	Showing customer ordered product of total amount
Profit	Showing company profit
Quantity	Showing how much quantity of product purchased by customer
Category	Showing the which category of product, the customer ordered.
Sub-category	Showing the which sub-category of product, the customer ordered

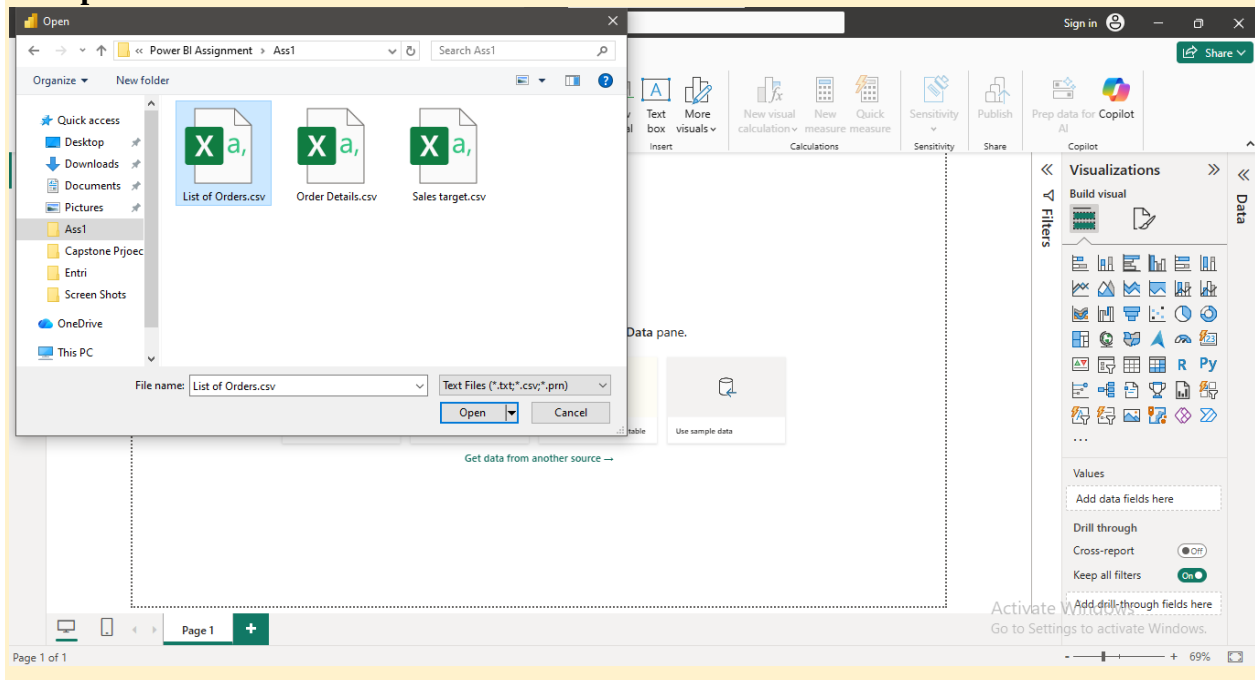
Sales Target Table:

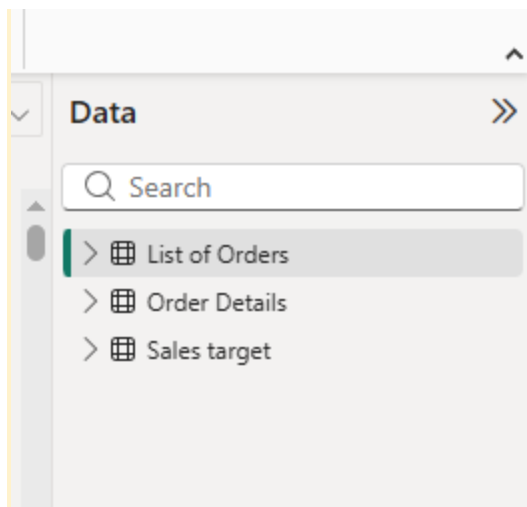
Column Name	Column Description
Month of Order Date	Showing customer ordered the product of month
Category	Showing the which category of ordered product
Target	Showing the target amount.

Data Transformation (ETL):

1.Data Import:

a. Import List of Orders.csv into Power BI.





b. Open the list of orders in the Power Query Editor using the Transform Data option.

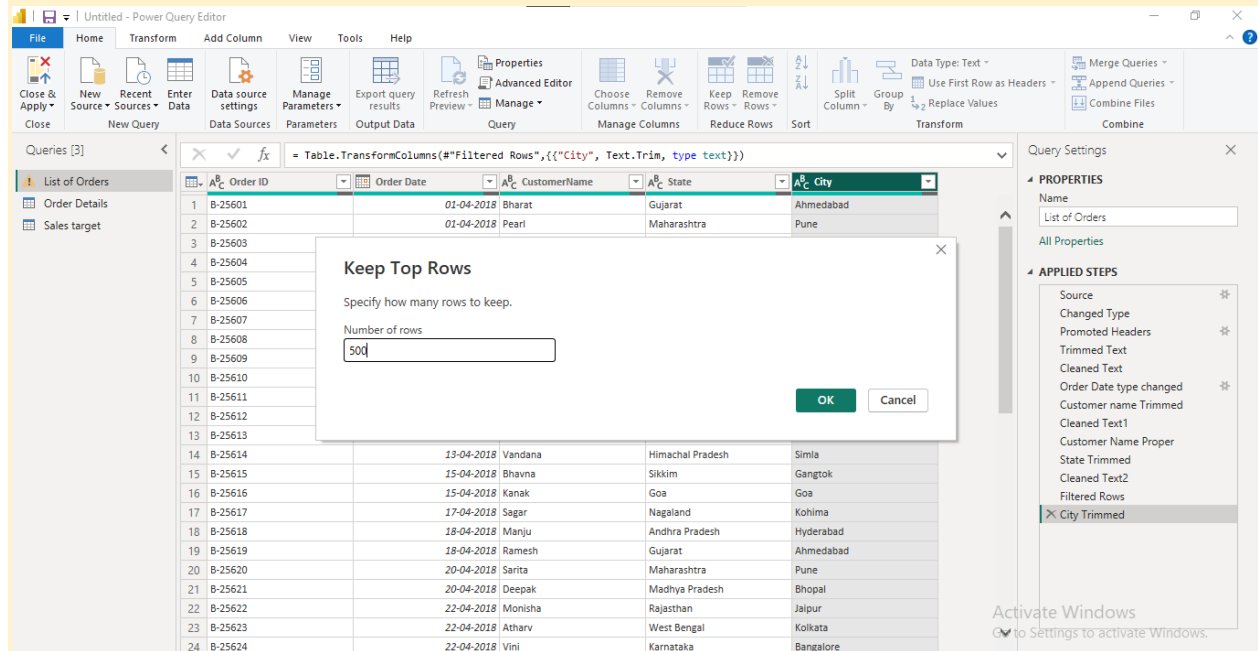
The screenshot shows the Power Query Editor with the 'List of Orders' query loaded. The main workspace displays a table with the following data:

	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal

The 'APPLIED STEPS' pane on the right shows the following steps: Source, Changed Type, Promoted Headers, and Changed Type1. The 'Properties' pane on the right shows the 'Name' property set to 'List of Orders'.

2. Row Limiting & Data Types:

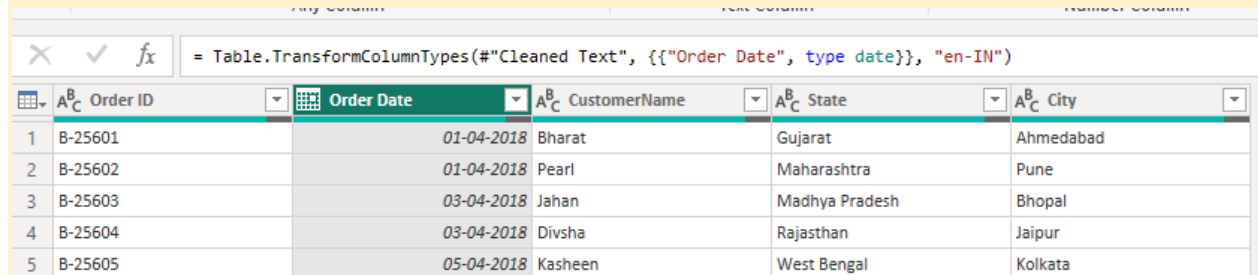
a. Restrict the List of Orders table to the first 500 rows:



Power Query Editor interface showing the 'Keep Top Rows' dialog box. The dialog prompts to 'Specify how many rows to keep' and shows 'Number of rows' set to 500. The background table displays order data with columns: Order ID, Order Date, CustomerName, State, and City.

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603				
B-25604				
B-25605				
B-25606				
B-25607				
B-25608				
B-25609				
B-25610				
B-25611				
B-25612				
B-25613				
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
B-25623	22-04-2018	Atharv	West Bengal	Kolkata
B-25624	22-04-2018	Vini	Karnataka	Bangalore

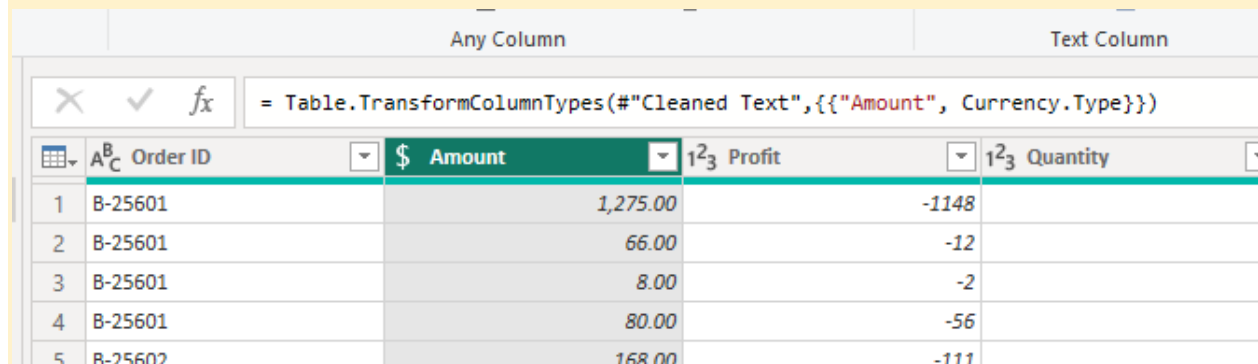
b. Set the Order Date column to the Date data type:



Power Query Editor interface showing the formula bar with the M code: `= Table.TransformColumnTypes("#Cleaned Text", {{"Order Date", type date}}, "en-IN")`. The background table displays order data with columns: Order ID, Order Date, CustomerName, State, and City.

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata

c. Change the Amount and Target columns to fixed decimal numbers:



Power Query Editor interface showing the formula bar with the M code: `= Table.TransformColumnTypes("#Cleaned Text", {{"Amount", Currency.Type}})`. The background table displays order data with columns: Order ID, Amount, Profit, and Quantity.

Order ID	Amount	Profit	Quantity
B-25601	1,275.00	-1148	
B-25601	66.00	-12	
B-25601	8.00	-2	
B-25601	80.00	-56	
B-25602	168.00	-111	

3. Text Formatting:

a. Format the Customer Name column to proper case to ensure consistent capitalization:

= Table.TransformColumns("#Cleaned Text1",{{"CustomerName", Text.Proper, type text}})

	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore

4. Column Creation:

a. Merge the City and State columns to create a new column named Location in the format: City, State:

Structure		Formatting		Properties	
		1 Location = 'List of Orders'[City]& " - "&'List of Orders'[State]			
Order ID	Order Date	CustomerName	State	City	Location
B-25627	23 April 2018	Hitika	Madhya Pradesh	Indore	Indore - Madhya Pradesh
B-25637	26 April 2018	Ashmi	Madhya Pradesh	Indore	Indore - Madhya Pradesh
B-25645	01 May 2018	Yaanvi	Madhya Pradesh	Indore	Indore - Madhya Pradesh
B-25651	07 May 2018	Anurag	Madhya Pradesh	Indore	Indore - Madhya Pradesh
B-25655	11 May 2018	Nida	Madhya Pradesh	Indore	Indore - Madhya Pradesh

b. Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage):

Structure

Formatting

Properties

✕

✓

1 Profit Margin % = `DIVIDE('Order Details'[Profit],'Order Details'[Amount])`

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin %
B-25602	561	212	3	Clothing	Saree	38%
B-25602	119	-5	8	Clothing	Saree	-4%
B-25603	193	-166	3	Clothing	Saree	-86%
B-25604	157	5	9	Clothing	Saree	3%
B-25605	75	0	7	Clothing	Saree	0%
B-25609	25	-5	4	Clothing	Saree	-20%
B-25610	43	0	3	Clothing	Saree	0%
B-25611	160	-59	2	Clothing	Saree	-37%

5. Conditional Column:

a. Add a conditional column named Profit Status based on:

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

1 Profit Status = IF('Order Details'[Profit]>0,"Profit",IF('Order Details'[Profit]<0,"Loss","Break-Even"))

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin %	Profit Status
B-25602	561	212	3	Clothing	Saree	38%	Profit
B-25602	119	-5	8	Clothing	Saree	-4%	Loss
B-25603	193	-166	3	Clothing	Saree	-86%	Loss
B-25604	157	5	9	Clothing	Saree	3%	Profit
B-25605	75	0	7	Clothing	Saree	0%	Break-Even
B-25609	25	-5	4	Clothing	Saree	-20%	Loss
B-25610	43	0	3	Clothing	Saree	0%	Break-Even
B-25611	160	-59	2	Clothing	Saree	-37%	Loss
B-25613	1603	0	9	Clothing	Saree	0%	Break-Even
B-25619	353	90	8	Clothing	Saree	25%	Profit

6. Merging Data (Joins):

a. Merge the List of Orders and Order Details tables using Order ID:

Power Query Editor - Merge dialog box configuration:

- Tables:** List of Orders, Order Details
- Join Kind:** Left Outer (all from first, matching from second)
- Fuzzy matching options:** The selection matches 500 of 500 rows from the first table.
- Query Settings:**
 - PROPERTIES:** Name: List of Orders
 - APPLIED STEPS:** Source, Changed Type, Promoted Headers, Trimmed Text, Cleaned Text, Order Date type changed, Customer name Trimmed, Cleaned Text1, Customer Name Proper, State Trimmed, Cleaned Text2, Filtered Rows, City Trimmed, Restrict 500 rows.

b. Name the merged output table as Orders Data:

= Table.ExpandTableColumn(Source, "Order Details", {"Order ID", "Amount", "Profit", "Quantity", "Category"},						
	Order ID	Order Date	CustomerName	State	City	Order Detail
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	B-25601
2	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	B-25601
3	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	B-25601
4	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	B-25601
5	B-25602	01-04-2018	Pearl	Maharashtra	Pune	B-25602
6	B-25602	01-04-2018	Pearl	Maharashtra	Pune	B-25602
7	B-25602	01-04-2018	Pearl	Maharashtra	Pune	B-25602
8	B-25602	01-04-2018	Pearl	Maharashtra	Pune	B-25602

Query Settings

PROPERTIES

Name

Orders Data (Merged Table)

All Properties

APPLIED STEPS

Source

Expanded Order Details

7. Handling Missing Data & Duplicate Data:

a. Removing Duplicate in List of Order table Order ID:

= Table.Distinct("#Removed Duplicates", {"Order ID"})						
	Order ID	Order Date	CustomerName	State	City	
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune	
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore	
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	
10	B-25610	09-04-2018	Yogesh	Bihar	Patna	
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok	
16	B-25616	15-04-2018	Kanak	Goa	Goa	
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima	
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune	
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata	
24	B-25624	22-04-2018	Vini	Karnataka	Bangalore	

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

Changed Type

Promoted Headers

Trimmed Text

Cleaned Text

Order Date type changed

Customer name Trimmed

Cleaned Text1

Customer Name Proper

State Trimmed

Cleaned Text2

Filtered Rows

City Trimmed

Restrict 500 rows

Removed Duplicates

Removed Duplicates1

b. Removing Duplicate in Order Details table Order ID:

Query Settings

Query Name: Order Details

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Trimmed Text
- Cleaned Text
- Amount type changed
- Removed Duplicates

	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1,275.00	-1,148.00	7	Furniture	Bookcases
2	B-25602	168.00	-111.00	2	Electronics	Phones
3	B-25603	1,355.00	-60.00	5	Clothing	Trousers
4	B-25604	65.00	17.00	2	Clothing	T-shirt
5	B-25605	75.00	0.00	7	Clothing	Saree
6	B-25606	87.00	4.00	2	Clothing	Shirt
7	B-25607	50.00	15.00	4	Clothing	Leggings
8	B-25608	1,364.00	-1,864.00	5	Furniture	Tables
9	B-25609	485.00	29.00	4	Electronics	Electronic Ge
10	B-25610	1,076.00	-38.00	4	Electronics	Printers
11	B-25611	160.00	-59.00	2	Clothing	Saree
12	B-25612	259.00	-55.00	2	Furniture	Chairs
13	B-25613	1,603.00	0.00	9	Clothing	Saree
14	B-25614	494.00	54.00	4	Furniture	Bookcases
15	B-25615	68.00	20.00	5	Clothing	Hankerchief
16	B-25616	42.00	12.00	5	Clothing	Hankerchief
17	B-25617	305.00	-270.00	5	Electronics	Electronic Ge
18	B-25618	362.00	127.00	1	Furniture	Bookcases
19	B-25619	353.00	90.00	8	Clothing	Saree
20	B-25620	193.00	46.00	1	Electronics	Printers
21	B-25621	233.00	-10.00	5	Electronics	Electronic Ge
22	B-25622	534.00	0.00	3	Clothing	Saree
23	B-25623	53.00	1.00	4	Clothing	Stole

7. Sorting and Filtering Data:

a. Sort records in the Orders Data table by Order Date in descending order:

Order ID	Order Date	CustomerName
B-26100	31 March 2019	Hitika
B-26099	30 March 2019	Bhishm
B-26098	29 March 2019	Pinky
B-26095	28 March 2019	Monisha
B-26097	28 March 2019	Vini

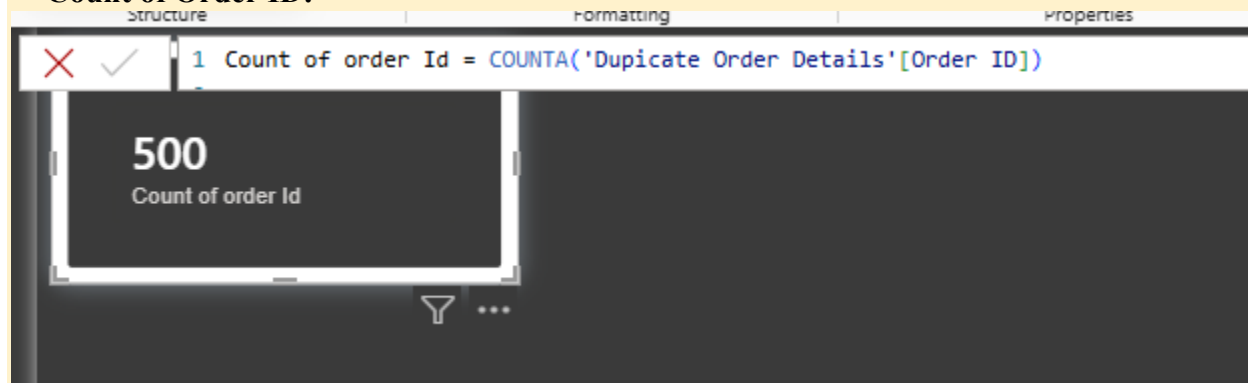
b. Apply filters to analyze orders from a specific state (e.g., Tamil Nadu):

Order ID	Order Date	CustomerName	State	City	Order Details.Order ID	Order Details.Amount	Order Details.Profit	Order Details.Quantity	Order Details.Category
B-26081	22 March 2019	Aarushi	Tamil Nadu	Chennai	B-26081	₹ 359	338 - ₹	5	Furniture
B-26018	14 February 2019	Aarushi	Tamil Nadu	Chennai	B-26018	₹ 326	₹ 107	3	Furniture
B-26008	09 February 2019	Kalyani	Tamil Nadu	Chennai	B-26008	₹ 22	₹ 4	1	Clothing
B-25860	15 November 2018	Akshay	Tamil Nadu	Chennai	B-25860	₹ 112	₹ 24	3	Clothing
B-25788	21 September 2018	Dinesh	Tamil Nadu	Chennai	B-25788	₹ 12	₹ 3	1	Clothing
B-25716	11 July 2018	Surabhi	Tamil Nadu	Chennai	B-25716	₹ 58	₹ 0	4	Clothing
B-25698	23 June 2018	Amisha	Tamil Nadu	Chennai	B-25698	₹ 87	83 - ₹	5	Clothing
B-25608	08 April 2018	Aarushi	Tamil Nadu	Chennai	B-25608	₹ 1,364	1,864 - ₹	5	Furniture

9. Grouping and Aggregating Data:

a. Duplicate the Order Details table and perform the following:

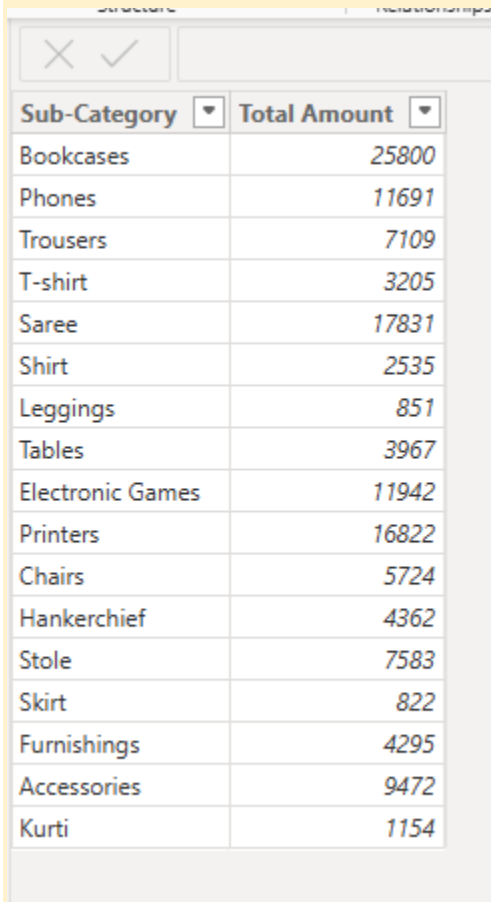
○ Count of Order ID:



○ Average Profit by Category:

Structure		Relationships	
☐	☒		
Category	Average Profit		
Furniture	20.44		
Electronics	-19.38		
Clothing	15.29		

○ **Total Amount by Sub-Category:**



The screenshot shows a window with a close button (X) and a checkmark button. Below the buttons is a table with two columns: 'Sub-Category' and 'Total Amount'. The table contains 17 rows of data. The values for 'Total Amount' are displayed in a red font.

Sub-Category	Total Amount
Bookcases	25800
Phones	11691
Trousers	7109
T-shirt	3205
Saree	17831
Shirt	2535
Leggings	851
Tables	3967
Electronic Games	11942
Printers	16822
Chairs	5724
Hankerchief	4362
Stole	7583
Skirt	822
Furnishings	4295
Accessories	9472
Kurti	1154

b. Duplicate the Sales Target table and aggregate the total Target by Month of order date:

1 Monthly Sales Target =

2 SUMMARIZECOLUMNS(

3 'Duplicate Sales Target'[Month Name],

4 "Total Target", SUM('Duplicate Sales Target'[Target]))

5)

6

Month Name	Total Target
April	₹ 31,400
May	₹ 31,500
June	₹ 31,600
July	₹ 33,800
August	₹ 33,900
September	₹ 34,000
October	₹ 36,100
November	₹ 36,300
December	₹ 36,400
January	₹ 43,500
February	₹ 43,600
March	₹ 43,800

10. Data Modeling:

a. Establish a relationship between:

- List of Orders and Order Details using Order ID:

← New relationship

Select tables and columns that are related.

From table

List of Orders

City	CustomerName	Location	Order Date	Order ID	State
Indore	Hitika	Indore - Mad...	23 April 2018	B-25627	Madhya Prad...
Indore	Ashmi	Indore - Mad...	26 April 2018	B-25637	Madhya Prad...
Indore	Yaanvi	Indore - Mad...	01 May 2018	B-25645	Madhya Prad...

To table

Order Details

Category	Order ID	Profit Mergin ...	Profit Status	Quantity	Sub-Category
Clothing	B-25623	2%	Profit	4	Stole
Clothing	B-25627	-0.709090909...	Loss	4	Stole
Clothing	B-25630	0.0902255639...	Profit	5	Stole

Cardinality

Many to many (*:*)

Cross-filter direction

Both

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

✖ There's already a relationship between these two columns.

- Order Details and Sales Target using Category:

← New relationship



Select tables and columns that are related.

From table

Order Details

Category	Order ID	Profit Mergin ...	Profit Status	Quantity	Sub-Category
Clothing	B-25623	2%	Profit	4	Stole
Clothing	B-25627	-0.709090909...	Loss	4	Stole
Clothing	B-25630	0.0902255639...	Profit	5	Stole

To table

Sales target

Category	Month of Ord...
Furniture	18-04-2026
Furniture	18-05-2026
Furniture	18-06-2026

Cardinality

Many to many (*:*)

Cross-filter direction

Both

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

✖ There's already a relationship between these two columns.

b. Use Manage Relationships to ensure relationships are active and correctly defined:

Manage relationships

+ New relationship

Autodetect

Edit

Delete

Filter

☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ Duplicate Order Details (Categ...		Group By Order Details (Categ...	Inactive	...
☐ Duplicate Order Details (Order ...		Order Details (Order ID)	Active	...
☐ Duplicate Order Details (Sub-C...		Group By Sub-category Order ...	Inactive	...
☐ List of Orders (Order ID)		Order Details (Order ID)	Active	...
☐ Order Details (Category)		Sales target (Category)	Active	...
☐ Order Details (Category)		Group By Sub-category Order details (Sub-Category)	Inactive	...
☐ Order Details (Sub-Category)		Group By Sub-category Order ...	Active	...
☐ Orders Data (Merged Table) (O...		Order Details (Order ID)	Active	...
☐ Sales target (Category)		Group By Order Details (Categ...	Inactive	...

Act

Close

Conclusion:

The data transformation and modeling workflow successfully converts three separate raw e-commerce data streams into a cohesive, analytics-ready data model within Power BI. By executing targeted ETL (Extract, Transform, Load) procedures, the project resolves key operational reporting challenges and prepares the foundation for strategic business insights.